

Best-Conditioned Eigenvectors of Finite Compressed Shifts: an Exact Diagonal SDP and the Regular-Polygon Formula

Candidate solution packet for arXiv:2006.04901, Question 1.1

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Status. Candidate major partial result; likely valid; pending human review. The arbitrary-zero problem receives an exact semidefinite optimization characterization. Closed formulas are proved in degree two and for all regular pseudohyperbolic polygons. No elementary closed formula for a generic zero configuration is claimed.

1 Source question

The source is Bickel, Gorkin, Greenbaum, Ransford, Schwenninger, and Wegert, *Crouzeix's Conjecture and Related Problems*, arXiv:2006.04901 [1]. If Θ is a finite Blaschke product with distinct zeros $z_1, \dots, z_n \in \mathbb{D}$, the compressed shift S_Θ has an $n \times n$ matrix representation M_Θ . For a matrix A with distinct eigenvalues, the source defines

$$\eta(A) = \inf\{\|X\|_2\|X^{-1}\|_2 : A = X\Lambda X^{-1}, \Lambda \text{ diagonal}\}.$$

Question 1.1 on source page 4 asks:

What is the minimal condition number of an eigenvector matrix X of M_Θ for a general finite Blaschke product Θ with distinct zeros?

For a matrix A with distinct eigenvalues, one can similarly define the minimal condition number of an eigenvector matrix of A by

$$\eta(A) := \inf\{\|X\|\|X^{-1}\| : A = X\Lambda X^{-1}\}. \quad (5)$$

and as in (4), if $\eta(A) \leq 2$, then Crouzeix's conjecture holds for A . In Subsection 2.2, we study this setup for matrices M_Θ that are representations of the compression of the shift S_Θ , associated with a finite Blaschke product Θ with distinct zeros in $\mathbb{D}$. First, in Theorem 2.4, we provide tractable formulas for specific X and X^{-1} . Then, in Theorem 2.6, we use these formulas to obtain a bound on $\eta(M_\Theta)$ in terms of the separation of the zeros of Θ . Because compressions of the shift are quite important, we pose the following question:

Question 1.1. *What is the minimal condition number of an eigenvector matrix X of M_Θ for a general finite Blaschke product Θ with distinct zeros?*

Figure 1: The definition of η and Question 1.1 on source page 4.

2 Statement of the result

For $z_1, \dots, z_n \in \mathbb{D}$, let $G = G(z_1, \dots, z_n)$ be the normalized Szegő-kernel Gram matrix

$$G_{ij} = \frac{\sqrt{(1 - |z_i|^2)(1 - |z_j|^2)}}{1 - \bar{z}_i z_j}. \quad (1)$$

It is positive definite when the points are distinct. With the opposite inner-product convention, the column Gram matrix is $\bar{G}$ instead; complex conjugation preserves all eigenvalues and the Loewner inequalities below because the comparison matrices are real diagonal. For positive Hermitian matrices B, C , write $B \preceq C$ for the Loewner order, and write $\kappa_2(B) = \lambda_{\max}(B)/\lambda_{\min}(B)$.

Theorem 1 (Exact diagonal SDP). *Let Θ be a finite Blaschke product with distinct zeros $z_1, \dots, z_n$. Then*

$$\eta(M_\Theta)^2 = \min_{D > 0 \text{ diagonal}} \kappa_2(DGD) \quad (2)$$

$$= \min \{t \geq 1 : \text{there is a positive diagonal } P \text{ with } P \preceq G \preceq tP\}. \quad (3)$$

Both minima are attained. Thus $\eta(M_\Theta)^2$ is the optimum of a one-parameter quasiconvex semidefinite feasibility problem depending only on the zero set, not on its ordering.

Let

$$\rho(z, w) = \left| \frac{z - w}{1 - \bar{z}w} \right|$$

denote pseudohyperbolic distance.

Theorem 2 (Degree two and a pairwise lower bound). *If $n = 2$, put $\rho = \rho(z_1, z_2)$ and $c = \sqrt{1 - \rho^2}$. Then*

$$\eta(M_\Theta) = \sqrt{\frac{1+c}{1-c}} = \frac{1+c}{\rho}. \quad (4)$$

For arbitrary n , the sharp pairwise obstruction is

$$\eta(M_\Theta) \geq \max_{i < j} \sqrt{\frac{1+c_{ij}}{1-c_{ij}}}, \quad c_{ij} = \sqrt{1 - \rho(z_i, z_j)^2}. \quad (5)$$

Call a zero set a *regular pseudohyperbolic n -gon of radius r* if it is the image under one disk automorphism of

$$\{r, r\omega, \dots, r\omega^{n-1}\}, \quad \omega = e^{2\pi i/n}, \quad 0 < r < 1.$$

Theorem 3 (Regular-polygon formula). *If the zeros of Θ form a regular pseudohyperbolic n -gon of radius r , then*

$$\boxed{\eta(M_\Theta) = r^{-(n-1)}}. \quad (6)$$

Equal norms for the eigenvectors are optimal; no unequal column scaling can improve the condition number.

The source observes that $\eta(A) \leq 2$ is sufficient for Crouzeix's conjecture for A . Theorem 3 therefore makes that test exact for this family:

Corollary 4. *For a regular pseudohyperbolic n -gon, the condition-number test $\eta(M_\Theta) \leq 2$ holds exactly when*

$$r \geq 2^{-1/(n-1)}.$$

This is a statement about the sufficient test, not a failure assertion for Crouzeix's conjecture below the threshold.

3 Idea of the proof

The source diagonalizes M_Θ using an explicit triangular matrix, but that formula depends on an ordering of the zeros and obscures the minimization over eigenvector lengths. Passing to the adjoint removes both issues. The eigenvectors of S_Θ^* are precisely the normalized Szegő kernels at the zeros, so every choice of eigenvector matrix differs only by diagonal column scaling. Its squared singular values are the eigenvalues of DGD . This gives the exact diagonal-scaling problem, and a congruence turns it into the SDP in (3).

For a regular polygon, G is circulant and its Fourier eigenvectors are flat: every coordinate has modulus $n^{-1/2}$. Pull the largest- and smallest-eigenvalue Fourier vectors back by D^{-1} . Their Rayleigh quotients for DGD have exactly the same denominator, regardless of D . Their quotient therefore certifies that diagonal scaling cannot beat $\kappa_2(G)$. A geometric-series computation of the Fourier eigenvalues then gives $r^{-2(n-1)}$ for the squared condition number.

4 Proofs

4.1 Kernel eigenvectors and the diagonal SDP

Let

$$k_z(w) = \frac{1}{1 - \bar{z}w}, \quad e_z = (1 - |z|^2)^{1/2} k_z.$$

If $\Theta(z_j) = 0$, then $k_{z_j} \in K_\Theta = H^2 \ominus \Theta H^2$ because $\langle \Theta h, k_{z_j} \rangle = (\Theta h)(z_j) = 0$. Since K_Θ is backward-shift invariant,

$$S_\Theta^* k_{z_j} = S^* k_{z_j} = \bar{z}_j k_{z_j}.$$

The eigenvalues are distinct, so these n kernel vectors form an eigenbasis of the n -dimensional model space.

If $A = X\Lambda X^{-1}$, then $A^* = X^{-*}\Lambda^*X^*$ and $\kappa_2(X^{-*}) = \kappa_2(X)$. Applying the same observation to A^* gives

$$\eta(A) = \eta(A^*). \quad (7)$$

Unitary changes of basis also preserve η . It follows that every eigenvector matrix relevant to $\eta(M_\Theta)$ can be represented by columns $d_j e_{z_j}$, where $d_j > 0$: phases and permutations have no effect on the singular values. Its Gram matrix is DGD or $D\bar{G}D$, according to inner-product convention, with $D = \text{diag}(d_1, \dots, d_n)$. The two matrices have the same eigenvalues, so we use DGD . Since

$$\kappa_2(X)^2 = \kappa_2(X^*X),$$

this proves (2).

Fix $t \geq 1$. The inequality $\kappa_2(DGD) \leq t$ holds if and only if there is an $s > 0$ such that

$$sI \preceq DGD \preceq t sI.$$

Congruencing by D^{-1} gives

$$sD^{-2} \preceq G \preceq t sD^{-2}.$$

Thus $P = sD^{-2}$ is feasible in (3). Conversely, if $P \preceq G \preceq tP$ for a positive diagonal P , then $D = P^{-1/2}$ gives

$$I \preceq DGD \preceq tI.$$

This proves equality of the two optima.

For completeness, the second optimum is attained. It is feasible at $t_0 = \kappa_2(G)$ by taking $P = \lambda_{\min}(G)I$. If $t \leq t_0$ is feasible, then comparison of diagonal entries in $P \preceq G \preceq tP$ and $G_{ii} = 1$ gives

$$t_0^{-1} \leq p_i \leq 1.$$

The relevant feasible set is therefore compact, and the Loewner inequalities are closed. This proves Theorem 1.

4.2 Automorphisms of the disk

We record the invariance needed below. It suffices to consider

$$\phi(z) = e^{i\theta} \frac{z - a}{1 - \bar{a}z}.$$

The identities

$$\begin{aligned} 1 - |\phi(z)|^2 &= \frac{(1 - |a|^2)(1 - |z|^2)}{|1 - \bar{a}z|^2}, \\ 1 - \overline{\phi(z_i)}\phi(z_j) &= \frac{(1 - |a|^2)(1 - \bar{z}_i z_j)}{(1 - a\bar{z}_i)(1 - \bar{a}z_j)} \end{aligned}$$

show that

$$G(\phi(z_1), \dots, \phi(z_n)) = U^* G(z_1, \dots, z_n) U, \quad U_{jj} = \frac{1 - \bar{a}z_j}{|1 - \bar{a}z_j|}. \quad (8)$$

The diagonal unitary U commutes with every positive diagonal D , so (2) is invariant under applying one disk automorphism to all zeros.

4.3 Two points and the pairwise obstruction

For two points, a diagonal unitary changes G to

$$\begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}, \quad c = |G_{12}|.$$

A direct calculation gives

$$c^2 = \frac{(1 - |z_1|^2)(1 - |z_2|^2)}{|1 - \bar{z}_1 z_2|^2} = 1 - \rho(z_1, z_2)^2. \quad (9)$$

Let $D = \text{diag}(a, b)$. Overall scalar multiplication does not change a condition number, so normalize $ab = 1$. Then the determinant of DGD is $1 - c^2$, while its trace is $a^2 + b^2 \geq 2$, with equality exactly when $a = b = 1$. For a positive 2×2 matrix with fixed determinant, the ratio of its two eigenvalues is strictly increasing with its trace. Equal scaling is therefore optimal, and

$$\eta(M_\Theta)^2 = \frac{1 + c}{1 - c}.$$

Using $\rho^2 = 1 - c^2$ gives both forms in (4).

For general n and any positive diagonal D , the (i, j) principal block of DGD is a scaled two-point Gram matrix. Eigenvalue interlacing implies

$$\kappa_2(DGD) \geq \kappa_2((DGD)_{\{i,j\} \times \{i,j\}}) \geq \frac{1 + c_{ij}}{1 - c_{ij}}.$$

Taking square roots, then the maximum over pairs and the infimum over D , proves Theorem 2.

4.4 Regular polygons

By (8), it is enough to treat $z_j = r\omega^j$ for $j = 0, \dots, n-1$. Put $q = r^2$. Then

$$G_{ij} = \frac{1-q}{1-q\omega^{j-i}}, \quad (10)$$

so G is circulant. For $0 \leq \ell < n$, let

$$f_\ell = n^{-1/2}(1, \omega^{-\ell}, \dots, \omega^{-(n-1)\ell})^T.$$

Expanding $(1 - q\omega^k)^{-1}$ as a geometric series and using the root-of-unity filter gives

$$\begin{aligned} \lambda_\ell &= (1-q) \sum_{k=0}^{n-1} \frac{\omega^{-\ell k}}{1-q\omega^k} \\ &= n(1-q) \sum_{h=0}^{\infty} q^{\ell+hn} = \frac{n(1-q)q^\ell}{1-q^n}. \end{aligned}$$

Thus $Gf_\ell = \lambda_\ell f_\ell$, with largest eigenvalue λ_0 and smallest eigenvalue λ_{n-1} .

It remains to prove that a diagonal scaling cannot reduce their ratio. Fix a positive diagonal $D = \text{diag}(d_0, \dots, d_{n-1})$, put $H = DGD$, and set $x_\ell = D^{-1}f_\ell$. Because all Fourier coordinates have modulus $n^{-1/2}$,

$$\|x_\ell\|^2 = \frac{1}{n} \sum_{j=0}^{n-1} d_j^{-2} =: \alpha_D \quad \text{for every } \ell.$$

Moreover,

$$\frac{x_\ell^* H x_\ell}{\|x_\ell\|^2} = \frac{f_\ell^* G f_\ell}{\alpha_D} = \frac{\lambda_\ell}{\alpha_D}.$$

The variational characterization of extreme eigenvalues now yields

$$\lambda_{\max}(H) \geq \frac{\lambda_0}{\alpha_D}, \quad \lambda_{\min}(H) \leq \frac{\lambda_{n-1}}{\alpha_D}.$$

Consequently

$$\kappa_2(DGD) \geq \frac{\lambda_0}{\lambda_{n-1}} = q^{-(n-1)} = r^{-2(n-1)}.$$

Equality holds when D is scalar. Taking square roots proves Theorem 3; Corollary 4 is immediate.

5 Verification and limitations

Independent checks. The accompanying script constructs the kernel Gram matrices directly and optimizes over logarithmic diagonal scales. For $2 \leq n \leq 8$ and $r \in \{0.2, 0.5, 0.8\}$, it reproduces $r^{-(n-1)}$ to numerical precision and returns scalar optimal scaling. It also checks the degree-two expression on 100 random pairs and verifies disk-automorphism invariance. A generic five-point example exhibits a strict improvement from unequal scaling, confirming that the regular-polygon argument is genuinely special. These computations are sanity checks, not ingredients of the proof.

Scope. Equation (3) is an exact general characterization and supports certified numerical computation by bisection in t . It does not simplify the answer for an arbitrary nonsymmetric zero set to a scalar expression in the pairwise pseudohyperbolic distances. Random tests show that simply taking $D = I$ is false as a general optimality rule. Thus the broad closed-form part of Question 1.1 remains open outside the solved families.

Bounded novelty check. On 12 August 2026, searches covered the run’s registry, solution, attempt, and proof-gap indexes; exact phrases involving “minimal condition number”, “finite Blaschke product”, “compressed shift”, and M_Θ ; the OpenAlex-indexed citation descendants of [1]; and the most relevant later primary sources found in that search, including arXiv:2306.12183 [2], arXiv:2312.04537 [3], and arXiv:2506.15444 [4]. No exact diagonal-SDP reduction or regular-polygon formula was located. This is a bounded search, not a proof of novelty; novelty confidence is medium.

Human-review focus. The most useful checks are: (i) the passage from M_Θ to the kernel eigenvectors of S_Θ^* and equality (7); (ii) the congruence equivalence in Theorem 1; (iii) the diagonal-unitary factor in (8); and (iv) the common-denominator Rayleigh quotient that proves optimality for the regular polygon.

References

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- [2] R. O’Loughlin and J. Virtanen, *Crouzeix’s conjecture for classes of matrices*, Linear Algebra Appl. **697** (2024), 277–292; [arXiv:2306.12183](#).
- [3] K. Bickel, G. Corbett, A. Glenning, C. Guan, and M. Vollmayr-Lee, *Crouzeix’s conjecture, compressions of shifts, and classes of nilpotent matrices*, Concrete Operators **11** (2024); [arXiv:2312.04537](#).
- [4] A. Renard, *Contractive upper triangular matrices with prescribed diagonals and superdiagonals*, Linear Algebra Appl. **724** (2025), 83–95; [arXiv:2506.15444](#).